Manthan Sontakke

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PROFESSIONAL SUMMARY

ECE undergraduate (3rd year) with hands-on experience in embedded C/C++, RISC-V SoC design and memory-mapped peripherals (UART, SPI, GPIO, timers, interrupts). Built an RTL-to-GDSII SoC and a bare-metal driver flow on Zynq with AMD Vitis/Vivado. Looking for an embedded software internship of 6+ months at the hardware/software boundary.

EDUCATION

B.Tech, Electronics and Communication Engineering, Vellore Institute of Technology, Bhopal — Aug 2023 - May 2027 | CGPA: 8.66/10

TECHNICAL SKILLS

- **Languages & Scripting:** C (C99), C++ (C++14), Embedded C, Python, TCL (build automation), Bash/Linux scripting
- **Embedded Software:** Bare-metal firmware, drivers for memory-mapped peripherals, interrupt handling, ESP32, Git
- **FPGA / SoC:** AMD Vitis, Vivado, Zynq-7000 (ARM Cortex-A9), AXI4-Lite, SystemVerilog, Verilog, RISC-V (RV32I), Pipelining
- **Protocols:** UART, SPI, I2C, AXI, Wishbone, GPIO, PWM
- **Tools:** Yosys, OpenLane, Icarus Verilog, Verilator, cocotb, LTspice, MATLAB

PROJECTS

MiniSoC-32: RISC-V Based System-on-Chip (RTL to GDSII) | GitHub — Oct 2026

SystemVerilog, Yosys, OpenLane, SkyWater 130nm

- Designed a 32-bit SoC with a 5-stage pipelined RV32I core (forwarding, load-use stall, branch flush), a Wishbone bus, and memory-mapped UART, SPI, GPIO and timer peripherals with interrupt support.
- Verified with 13 self-checking testbenches and firmware-driven gate-level simulation; diagnosed a synthesis bug where an unloaded ROM caused Yosys to prune the design (140 cells to 34k).
- Implemented RTL-to-GDSII in OpenLane/sky130 (about 34k cells, 1.19 mm^2), closing timing at 33 MHz across all 9 PVT corners with clean DRC, LVS and antenna checks.

Zynq-7000 AXI-Lite Peripheral with Bare-Metal Driver — Nov 2026 - In progress

Vivado, Vitis, C, TCL, AXI4-Lite

- Building a custom AXI4-Lite peripheral with control/status registers in the Programmable Logic and connecting it to the ARM Cortex-A9 processing system in Vivado.
- Developing a bare-metal C driver and test application in Vitis to configure the peripheral, handle its interrupt, and log results over UART.
- Automating Vivado project creation and bitstream generation with a TCL script, with the project version-controlled in Git.

QUENCH - IoT Smart Water Dispenser — Feb 2025 - Apr 2025

Embedded C/C++, ESP32, Sensors

- Engineered board design for a contactless IoT dispensing system, integrating an ESP32 and managing power distribution.
- Achieved response time under 2s and 95% flow measurement accuracy through precise sensor calibration.

EXPERIENCE

Embedded System Design Intern, Maven Silicon — Jun 2026 - Aug 2026

- Completed a structured curriculum covering digital design fundamentals, core microcontroller architecture, and GPIO.
- Executed system-level unit and integration tests on sensor inputs and digital control logic.

AI/ML Intern, MPOnline Limited — May 2026 - Jul 2026

- Developed data preprocessing pipelines and trained deep learning models using Python and TensorFlow.
- Deployed an end-to-end AI capstone project utilizing Git-based version control and MLOps practices.

CERTIFICATIONS

- **VLSI:** Core Fundamentals of Digital Design for VLSI Chip Design (L&T EduTech / Coursera)
- **NPTEL:** Certified in IoT (88%); Applied Machine Learning in Python (88.6%)